

Stock Market Prediction Using Machine Learning Techniques

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ABSTRACT

The prediction of stock market prices holds significant importance within the contemporary economic landscape. Consequently, there has been a notable surge in scholarly interest directed towards exploring novel avenues for enhancing stock market prediction capabilities. Recent research endeavors have illuminated the potential predictive value inherent in various data streams, including historical stock data and user-generated content sourced from platforms such as Twitter and web news. These investigations have revealed a discernible relationship between social mood, as reflected in online discourse, and future stock price movements. However, prior studies have often overlooked the incorporation of such sentiment-derived information, thus presenting an information gap. In the present study, we address this gap by proposing an effective methodology for the integration and analysis of multiple information sources to facilitate more accurate stock price predictions. Leveraging Long Short-Term Memory Recurrent Neural Network (LSTM-RNN) models, we conduct a comprehensive analysis of disparate data sources. Furthermore, we employ an ensemble approach, incorporating Weighted Average and Differential Evolution techniques, to refine the predictive accuracy of our models. Our findings demonstrate the efficacy of the proposed methodology in generating highly accurate stock price predictions across varying future time horizons, including one-day, seven-day, 15-day, and 30-day intervals. These predictions offer valuable insights for investors seeking to make informed decisions regarding their investment strategies and enable companies to gauge their anticipated performance within the stock market landscape.

Keywords: Stock Market Prediction; Sentiment Analysis; Neural Networks; Long-Short Term

I. INTRODUCTION

Financial markets are recognized as complex and dynamically evolving systems that exert a significant influence on the global economy. A myriad of macroeconomic factors, including political events, natural disasters, man-made crises, market sentiment, and direct market forces such as supply and demand dynamics, speculation, and expectations, collectively shape the behavior of financial stock markets. The deepening integration of globalization and financial markets exacerbates the intricacies inherent in predicting stock market movements solely through theoretical frameworks [1]. Consequently, it becomes increasingly challenging and impractical to rely solely on theoretical models for accurate predictions. Hence, it is imperative to incorporate the aforementioned exogenous factors into the prediction process.

Over the past two decades, the prediction of stock market movements has emerged as a prominent area of research activity. The proliferation of computational capabilities coupled with the abundance of big data resources has led researchers to increasingly rely on web-based sources for forecasting stock market behaviors. Machine learning techniques and Natural Language Processing (NLP) methods have become instrumental in this endeavor, offering avenues for analyzing and predicting stock market dynamics. Financial data, crucial for such predictions, manifest in various formats including text, videos, and images. Notably, textual data extracted from web news, search engine queries, and social media platforms has been extensively utilized by researchers in this domain. In this study, we propose an application aimed at predicting the stock return prices of 30 companies listed within the Dow Jones Industrial Average (DJIA) index. Our approach integrates sentiment analysis of Twitter data, text analysis of web news, search engine query metrics, historical stock prices, and employs a diverse array of machine learning techniques to develop a robust stock price return prediction algorithm. The efficacy of our

proposed algorithm is validated using real-world stock market data, ensuring its accuracy and reliability in practical applications.

Building upon the findings of prior research, which have established correlations between stock market data and the aforementioned data sources, we posit a strong potential for predicting stock return prices. Leveraging these predictive values offers valuable insights into how market dynamics influence the financial stability of individual companies. Subsequently, informed financial decisions can be made, potentially resulting in substantial cost savings amounting to millions of dollars.

Over the past two decades, the stock market has experienced significant commercialization, leading to heightened investor engagement. This trend has been facilitated by advancements in technology, affording investors substantial benefits through an expanded market scope.

In their research, S. Abdulsalam Sulaiman Olaniyi, Adewole Kayode S., and Jimoh R. G. highlight the exponential growth in the collection of various types of data, propelled by rapid advancements in computer technology. This surge in data generation and storage presents a significant opportunity for those capable of extracting valuable insights from the vast reservoirs of information [4].

Researchers are increasingly focusing their attention on the prediction of stock market values. As highlighted in [4], the study by David Enke and Suraphan Thawornwong underscores the existence of a relationship between historical data and future stock market values. To elucidate and exploit these predictive relationships, data mining techniques are commonly employed.

Hence, researchers have turned their attention towards Artificial Intelligence (AI) and Data Mining techniques. In the study referenced as [3], researchers collected daily stock value movements within a specified timeframe from the official daily listings of the Nigerian Stock Exchange. Subsequently, they constructed a comprehensive database to facilitate the

prediction of future stock values. Employing data mining methodologies, they developed a predictive tool utilizing regression analysis on time series data. This tool leveraged moving averages to forecast future stock market values.

Moreover, as articulated by Huina Mao, Scott Counts, and Johan Bollen in [5], stock market valuations are intricately linked to a wide array of both readily accessible and latent information, including behavioral and emotional factors such as social mood. Consequently, the analysis of social mood has emerged as a crucial and pivotal undertaking in the realm of stock market forecasting.

With the advent of advanced technology, stakeholders worldwide have increasingly utilized digital platforms to express their emotions and ideas concerning stocks [6]. In a study examining whether the daily volume of tweets could predict S&P 500 stock indicators, researchers investigated the predictive power of Twitter data. Another study focused on analyzing the contents of tweets, employing a textual analysis approach to discern relationships between mood indicators and the Dow Jones Industrial Average (DJIA).

The study referenced as [2] demonstrates a notable correlation between stock price returns and Twitter sentiments, suggesting its viability for further exploration. The researchers identified a relatively low Pearson's correlation and Granger causality between the respective time series. Utilizing a financial econometrics methodology known as event study, they scrutinized abnormal price returns observed during external events. This approach enabled them to predict stock price returns across a broader time series.

The advent and widespread adoption of the Internet during the 1990s heralded a significant transformation in the dissemination of news. Traditional newspaper articles, once confined to overnight printing on paper, became swiftly accessible in digital formats, catering to the expeditious demands of the financial market. Consequently, an interdisciplinary research field

emerged, dedicated to enabling computers to interpret news articles promptly and capitalize on opportunities within financial markets [7].

II. METHODS AND MATERIAL

The primary predictive variable in this study is the closed stock price return values of the Dow 30. The Dow Jones Industrial Average Index (DJI) is selected for its stability. To forecast stock price values, social media data, web news data, and search engine query data from January 1, 2016, to July 31, 2019, are utilized. Historical market data and real-time stock prices were acquired using the Yahoo Finance API.

Collected data needs to be preprocessed. Stock market were unavailable during the weekend. Hence null values were filled with the corresponding closed stock price of the Friday. There were multiple tweets per day for a company. All the tweets were preprocessed, and sentiment scores were calculated. Then sentiment mean score and the Twitter volume per day is calculated. s^- or s^+ is defined as the relative amount of polarity represented in a time interval.

$$S^+(t) = \frac{1}{n} \sum_{i=1}^n (S_i | S_i > 0) | (t)$$

In the provided equation, " s " represents the sentiment values per day, " t " signifies time, and " n " denotes the number of days. The sum of (s^-) and (s^+) is computed as the mean sentiment value for each given day. In instances where there were missing values for certain days in the web news dataset, the sentiment mean score for each day is calculated, and any missing values are filled using linear interpolation.

Given the inherent time series nature of stock prediction, a Recurrent Neural Network (RNN) is employed. Initially, a range of algorithms including LSTM (Long Short-Term Memory) and the ARIMA method were tested. LSTM exhibited superior performance among these algorithms. At the time, LSTM was considered a state-of-the-art RNN for time series prediction tasks. The input variables for both

multivariate and univariate time series forecasting included sentiment score (ts), web news sentiment (ns), Google Trends hits volume (gv), and closed stock price (c).

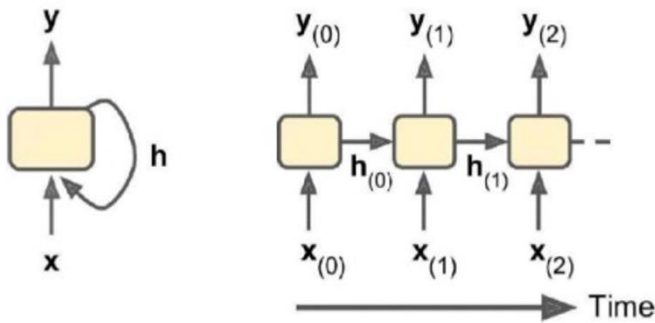
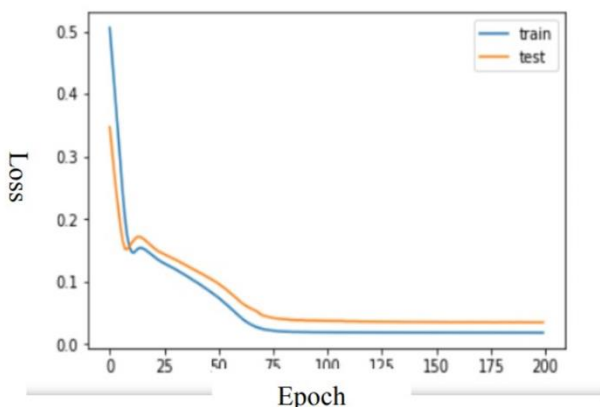


Figure 1: LSTM architecture

Each model is fitted with the relevant input data, and predictions are generated using the models that exhibit the best fit. These selected models are further fine-tuned to optimize their predictive accuracy.

III.RESULTS AND DISCUSSION

The predicted stock values are compared against the actual values to assess the accuracy of the forecasting model. The correlation between Twitter sentiment and stock market values of \$AAPL is reported as 0.5. The loss graph is plotted in figure 2.



Below is the time series analysis forecasting diagram for Twitter sentiment over a 60-day timestamp. The Root Mean Squared Error (RMSE) for this forecast is 0.013, indicating a relatively low level of prediction error. The accuracy of the forecast is reported as 0.962, reflecting a high level of precision in predicting Twitter sentiment over the specified time period.

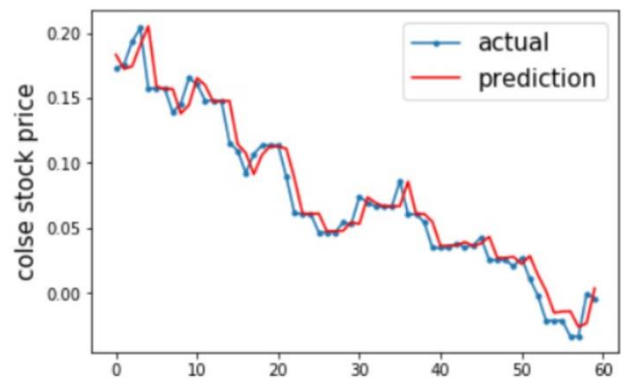


Figure 3: prediction vs actual price

For one-day prediction, the individual accuracies were 0.98, 0.97, and 0.96, respectively, for the Twitter, web news, and search engine query models. These models were weighted with values of 0.5, 0.3, and 0.2, respectively. The best score achieved for the ensembled model, considering these weighted accuracies, was 0.99.

IV.CONCLUSION

The primary objective of this study is to forecast stock market values of the Dow Jones 30 by integrating four main components discussed throughout the paper. Despite relatively low correlations observed, the integrated model demonstrated success, providing a clear path towards enhancing accuracy. Modal weights were adjusted according to the prediction time range, with weights for Twitter sentiment analysis and web news analysis decreasing as the time range increased. Conversely, for shorter prediction time ranges, the weight for Twitter analysis increased, suggesting its effectiveness for short-term predictions. Future research endeavors will focus on constructing a finance-based sentiment corpus rather than relying on text blob sentiment analysis. Additionally, beta values for stock market data will be incorporated as feature inputs. Moreover, macroeconomic variables such as exchange rates and gold rates will be investigated to further enhance predictive capabilities.

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